

Construction Element Detection for Site Progress Verification

A cloud-only YOLOv8 pipeline for automated progress evidence

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EXECUTIVE SUMMARY

Site progress in the Gulf is still recorded by photograph and narrative, with a two- to seven-day gap between capture and record. The photograph is archived; the information inside it is not. ISO 19650 asks for information delivered into a common data environment in a structured form, and a JPEG in a folder does not meet that bar.

This project trains a YOLOv8 detector on five construction classes — **steelbar, brick, pvcpipe, scaffold and excavator** — so that a site photograph yields a structured detection record: image, time, zone, class, confidence, bounding box. Steelbar, brick and pvcpipe act as proxies for structural, masonry and MEP first-fix activity in a zone; scaffold and excavator report site condition — temporary works standing, plant active.

The system is deliberately scoped as a **triage tool**. It directs a person to the photographs worth examining. It does not replace the examination, and it has no safety function of any kind. The entire pipeline runs in the browser — Roboflow for data, Google Colab for training — with no local installation, so a third party can reproduce every result from the repository alone.

METHOD

A public dataset, *seungyeon/construction-site-km7bh* (700 images, CC BY 4.0), was forked into our own Roboflow workspace and re-split from 70/20/10 to the required 80/20. We added no images of our own: 25 candidates were sourced, 20 were watermarked stock previews with no usable licence and were rejected, and 5 were too few to matter. Seven first-party photographs are held back entirely as an unbiased test set.

Every label follows a written class-definition contract that specifies inclusions, exclusions, the one-box-per-visually-separable-group rule and a 20-pixel minimum size. The contract exists because without written rules, annotation drifts between people and the model learns the drift rather than the object. SAM-assisted labelling was evaluated: it accelerated geometry on isolated objects and failed on grouped and amorphous classes, because it automates the drawing and never the judgement.

Dataset images	700
Split	560 / 140 (80 / 20)
Model	yolov8s.pt
Epochs	30
Image size	640
Batch	16
Hardware	Tesla T4, 15360 MiB
Training time	6.4 min

RESULTS

Class	Precision	Recall	mAP@50	mAP@50-95
All classes	0.950	0.884	0.943	0.809
brick	0.984	1.000	0.995	0.932
excavator	0.969	1.000	0.995	0.883
pvcpipe	0.986	0.930	0.964	0.835
scaffold	0.955	0.875	0.979	0.858
steelbar	0.855	0.612	0.781	0.535

INTERPRETATION

1. **Validation says 0.943 mAP@50. Our own photographs say almost nothing.** Across seven held-out images the model produced one marginal correct detection, two confident wrong-class detections and four blanks — including both classes we had designated as controls at recall 1.000.
2. **The cause is scale, and it is testable.** Holding weights, image and confidence fixed and changing only the input resolution, a blockwork wall moves from scaffold 0.78 to brick 0.46. A model whose class prediction depends on resampling has learned a texture signature, not an object. Lowering confidence to 0.02 recovers nothing.
3. **The fix is data, not capacity.** Re-split by scene group so validation can detect this at all; add wide-field imagery and train multi-scale; add hard negatives for the two confusions actually observed. A larger model would raise the leaking number and change none of the seven held-out results.

ERROR ANALYSIS AND NEXT ITERATION

Three false positives and three false negatives are documented with filenames in *docs/error_analysis.md*. Five of the six are explained by one cause — objects recognised at the training set's scale and not at any other — and the sixth by an annotation-convention dispute. Three prioritised data improvements follow, each naming a concrete action and the metric it should move.

RISK: THE ASYMMETRY BETWEEN THE TWO ERRORS

A false negative is silent. Steelbar is placed, the model misses it, no record is created, and nobody investigates because nobody knows there is anything to investigate. The gap propagates into the programme unchallenged. **A false positive is loud.** A phantom record is contradicted by the next site walk. It costs a wasted verification trip and is self-correcting.

For progress logging, therefore, **recall is the priority**. For site-condition alerts the asymmetry reverses: repeated false alarms teach supervisors to ignore the system. **One model, two thresholds** — low for logging, high for alerting — both read off the precision-recall curve rather than left at the default.

LIMITATIONS

- Reported metrics are inflated by train/validation leakage; the held-out set confirms it independently. They measure memorisation, not generalisation.
- Predictions change class with input resolution, so `imgsz` must be pinned end to end and confidence is not comparable across distributions.
- Wide-field shots fail — which is how site progress photography is actually taken. Validated only on close, subject-filling images.
- No safety function: no people, no PPE, no hazards. Treating it as a safety system would reduce site safety, not improve it.
- Presence only — never quantity, spacing, cover or conformance, and never a payment application. A human reviews every output.

GOVERNANCE AND LICENSING

Artefact	Licence	Owned	Constraint
Code and notebooks	MIT	Yes	None
Base dataset	CC BY 4.0	No	Attribution; referenced, not redistributed
Held-out test images	CC BY 4.0	Yes	Our own photographs; never trained on
Trained weights	AGPL-3.0	Derived	Commercial use needs AGPL compliance or a licence

No class detects or tracks people — by design: a person or PPE class would make this EU AI Act high-risk.